

EquiTie

Senior Software Engineer Case Study

Build the Investor Assistant: a personalised AI assistant for EquiTie investors

Timebox: 2 to 3 hours on the build. Format: a working prototype, a short write-up, and a build roadmap.

This case study is about judgment, not how much code you can write from scratch. We want to see how you use AI, how fast you move, your product sense, your verification discipline, and whether you can build leverage for an investment team. Work only from the dataset we provide.

1. Context

At EquiTie we are building the operating system for a modern venture-capital firm: deals, legal document workflows, investor operations, portfolio data, valuation tracking, and AI-assisted research, in one platform.

Our investors need to understand their position with the firm: the companies and deals they are in, their commitments and capital contributions, the current valuation and share price of each holding, their MOIC, upcoming management fees and capital calls, distributions, and their account statement. Today they get this from static statements and by emailing their relationship manager, then waiting. That does not scale, and it does not let an investor ask a question and get a grounded answer.

We want an AI assistant an investor can ask, in plain language, about their own portfolio. It should answer accurately, cite where the numbers come from, and adapt to the individual investor.

2. The Challenge

Build a working prototype of the EquiTie Investor Assistant: a conversational assistant that answers an investor's questions about their own portfolio, using the dataset we provide.

Assume the investor is already authenticated. You will be told which investor_id is logged in. The assistant must answer only about that investor, and must never expose another investor's data.

Questions it should handle

- Portfolio overview: holdings, current value, total committed vs contributed, and MOIC.
- A single position: current value, cost basis, and the share price the investor paid, including when they hold the same company across multiple rounds.
- Obligations: upcoming and overdue management fees, and upcoming capital calls.
- Realised outcomes: distributions and exits, and what was actually received after the performance fee (carry).
- Fees: what an investor pays on a given deal, reading the deal's standard schedule plus any discount on their allocation.
- Valuations: how a company's valuation has moved over time and what that does to the investor's MOIC.
- Account statement: a plain-language summary of capital contributions, fees, and distributions.

Personalisation (required)

The assistant should tailor each answer to the investor it is serving, using the profile fields in the data (age, tech savviness) plus signals it can derive, such as how many deals they are in and the sectors they invest in most. For example:

- A less technical or older investor gets plain language, jargon explained (what MOIC or carry means), and shorter answers.
- A sophisticated investor who is in many deals gets concise, data-dense answers that assume fluency.
- Where it helps, the assistant reflects the investor's portfolio shape, such as their most-active sectors or concentration, instead of answering generically.

Keep it professional and never patronising. Personalisation changes tone, depth, and framing. The underlying numbers stay correct and identical for everyone.

The dataset guide (Dataset Guide.md) describes the data model and the cases worth handling, including companies that appear in more than one round, per-investor share-price and fee discounts, several currencies, partial capital calls, exits, write-offs, down rounds, and similar company names.

3. Deliverables

1. **Working prototype.** A lightweight web app, notebook, CLI, or small repo, whatever best demonstrates the workflow. A polished UI is not required; a chat loop over the data is enough.
2. **README.** Architecture, which AI models or APIs you used and why, assumptions made, how to run it, and known limitations.
3. **ai-workflow.md (required).** See section 7.
4. **Build roadmap (required).** A short plan for the full product. See section 4.
5. **Walkthrough video (optional).** If you want, a short video (up to 5 minutes) showing the prototype in use and where it breaks.

4. The Roadmap

The prototype is the first step. We also want your plan for the real thing. Suppose you have six months and effectively unlimited budget to build a full relationship-manager bot inside the EquiTie iOS investor app: an assistant that does much of what a human relationship manager does today, not just answer questions. Set out how you would build it, in a short document (about 1 to 3 pages, or a few slides).

Cover at least:

- **Scope and capabilities.** What the full bot does beyond Q&A (for example proactive nudges, capital-call and fee reminders, document and KYC requests, onboarding, reporting, drafting investor comms), and what stays with a human.
- **Architecture and tech stack.** The end-to-end stack you would choose and why: client, backend, data layer, retrieval, model choice and hosting, orchestration, evaluation, observability, and security.
- **Data and integrations.** The systems it must connect to, such as the portfolio ledger, fund administration, CRM, KYC/AML, e-signature, comms, and valuation or market data, and how data flows between them.
- **AI approach and safety.** Models, grounding and retrieval, tool use, where deterministic code does the work instead of the model, how you evaluate quality, and guardrails and compliance (no investment advice, audit trail, data protection).

- **Team and hiring.** The team you would build: roles, seniority, headcount, and roughly when each hire lands over the six months.
- **Timeline.** A phased six-month plan with milestones and what ships at each phase.
- **Risks and cost.** The main risks, your build-versus-buy calls, and a rough cost shape.

We care about realistic, well-reasoned choices and clear trade-offs more than length or polish.

5. The Dataset

You will receive a folder of CSV files plus a dataset guide (Dataset Guide.md). Treat the report date as 25 June 2026 for anything upcoming or current. The core tables:

File	What it holds
investors.csv	112 investors: reporting currency, KYC, plus age and tech-savviness for personalisation.
portfolio_companies.csv	16 companies: sector, country, status (Active / Exited / Written Off).
deals.csv	21 deals, one per company-round SPV: currency, valuation, entry share price, contributed %, and the deal's standard fee schedule (mgmt / performance / structuring / admin).
allocations.csv	550 positions: commitment vs contributed, share-price discount, units, cost basis, and this investor's effective fees plus a fee_discount flag.
valuations.csv	Mark history per deal: share price and company valuation moving up and down; latest = current.
capital_calls.csv	Paid and upcoming capital calls (deal-level partial-call structure).
fees.csv	Management, structuring, and admin fees per allocation, at each investor's effective rate.
distributions.csv	Exit proceeds and secondary sales, gross and net of the performance fee.
statement_lines.csv	Per-investor account statement (deal currency, signed).
fx_rates.csv	Currency to USD as of the report date.

Work only from this dataset. Do not bring in external, real, or proprietary data.

6. Constraints and Guardrails

- Spend roughly 2 to 3 hours. Prioritise a narrow, correct, end-to-end flow over breadth or polish.
- Work only from the provided dataset. Do not bring in external or real-world data.
- Use any tools you like: Cursor, Claude, ChatGPT, Gemini, Copilot, Replit, Vercel, LangChain, LlamaIndex, MCP servers, local models, any stack.
- Do not build authentication, a landing page, or production infrastructure unless it is essential to your workflow.
- It is fine if the prototype is rough. It is not fine if it is ungrounded, impossible to run, or presented with false confidence.

7. ai-workflow.md (required)

A short file answering:

- Which AI tools and models did you use, and for what?
- Roughly what percentage of the code was AI-generated?
- What did you reject or materially change from AI suggestions, and why?
- How did you verify the assistant's answers were correct?
- If you had an autonomous coding agent for another 8 hours, what would you point it at next?

8. Evaluation Rubric

Area	Weight	What strong looks like
AI-native workflow	20%	Uses AI aggressively but intelligently: prompting, iteration, verification, tool selection, not blind copy-paste.
Reliability & verification	20%	Correct numbers, cites source rows, handles uncertainty and the edge cases, avoids false confidence.
Product judgment	15%	Tight scope on what matters to an investor; sensible, genuinely useful personalisation; builds the part that creates leverage.
Roadmap & systems thinking	20%	Credible six-month plan: sensible stack and integrations, realistic team and hiring, clear phasing, honest risks and trade-offs.
Technical architecture	10%	Clean separation of input, data and retrieval, model reasoning, and output. Sensible model choices.
Communication	10%	Clear README and roadmap; can explain trade-offs, assumptions, and failure modes.
Speed & shipping	5%	A usable end-to-end flow within the timebox over over-engineering.

9. Submission Checklist

- Prototype link or repository (required)
- README (required)
- ai-workflow.md (required)
- Build roadmap (required)
- Walkthrough video, up to 5 minutes (optional)
- Notes on limitations and failure modes (recommended)

Build something narrow, correct, personalised, and explainable.